

Minhazul Islam

AI Engineer — Agentic AI, LLM Systems & Full-Stack Development

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Open to full-time AI/ML engineering roles — Hangzhou / Shanghai / Remote · 2026–2027 · currently on a student visa, eligible for work-permit conversion after graduation (Dec 2026) · English (fluent) · Mandarin Chinese (working proficiency) · Bengali (native)

SUMMARY

AI engineer with 2+ years building production agentic AI systems end-to-end — LangGraph multi-agent architectures, RAG infrastructure, Dockerized deployment, and streaming APIs. Solo-built EDIA, a 17-node decision-intelligence platform serving 19 petrochemical products in live forecast use, cutting decision latency from 90–478s to 30–45s with zero LLM-computed numbers reaching users. Also built Zhanlu, a multi-tenant governed agent runtime with a 144-tool registry, MCP integration, and Docker sandboxing. M.Eng. candidate at Zhejiang University with 2 peer-reviewed papers (ICEC 2025) and 2 conference presentations (HHME 2026 PCC).

TECHNICAL SKILLS

LLM & Agent Systems: LangGraph (17-node FSM), multi-agent orchestration, Model Context Protocol (MCP), sub-agent delegation, tool/skill registry design, LLM-as-a-judge evaluation, prompt & context engineering

RAG & Data Grounding: hybrid vector + SQL retrieval, ChromaDB, BAAI/bge-m3 embeddings, context budgeting, semantic memory, NL2SQL pipelines, output validation & guardrails

Backend & Infrastructure: FastAPI, SSE streaming, PostgreSQL, Redis, MySQL, MinIO, Docker, Nginx, systemd, JWT auth, Prometheus/Grafana, GitHub Actions CI/CD

ML & Forecasting: PyTorch, Transformers, XGBoost, ARIMA/LSTM/STL ensembles, walk-forward MAPE backtesting, local inference (vLLM, Ollama, Qwen), multi-provider model routing

Languages & Tools: Python (asyncio), C/C++, SQL, JavaScript, React, Git, Autodesk Inventor / Fusion 360 API

Research Methods: user studies (N=24 within-subjects), usability evaluation (SUS), qualitative interview studies, systematic reviews

EXPERIENCE

AI Infrastructure Engineer (Intern) — Synexia AI 2025 – 2026

Zhejiang, China · Solo-built two production AI platforms: EDIA (enterprise decision intelligence) and Zhanlu (governed multi-agent runtime).

EDIA — Enterprise Decision Intelligence Agent (17-node LangGraph FSM · 15 agents · 57 skills · 19 products)

- Architected a 17-node LangGraph state machine orchestrating 15 specialized agents and 57 atomic skills across 5 execution tiers — conditional routing, parallel branches, sequential quality gates, and persistent multi-turn session state.
- Eliminated numerical hallucination architecturally: a deterministic pre-rendering pipeline computes and locks every business-critical metric in Python before any LLM narration, with a validation engine enforcing exact-value compliance and strict source attribution.
- Cut end-to-end decision latency from 90–478s to 30–45s by fusing ChromaDB vector search with 38 SQL views mapped to 19 products — SQL round-trips dropped from 77 to 2–5 and fetched rows from 180K to ~200. Every result is an immutable checksummed snapshot cited by ID, so live database credentials never reach LLM context.
- Deployed multi-horizon forecasting (ARIMA + XGBoost + LSTM + STL ensemble) with walk-forward MAPE backtesting and an Honesty Gate that auto-downgrades to conservative forecasts when model uncertainty exceeds thresholds (best-case MAPE 1.4%).
- Operated the platform in production on FastAPI + Nginx + systemd with SSE streaming and JWT auth, behind a dual-LLM security architecture in which a local Qwen model sanitizes ERP data before any external provider receives a briefing.

Tech: LangGraph · FastAPI · ChromaDB · PyTorch · XGBoost · statsmodels · Docker · nginx · SSE · MySQL · Redis

Zhanlu — Enterprise AI Agent Infrastructure (144-tool registry · MCP client/server · Docker sandboxing)

- Built a multi-tenant runtime for governed, tool-using AI agents: permission-filtered dispatch over a 144-tool registry, MCP client/server integration with OAuth, provider-health tracking with circuit-breaking, fallback routing, and encrypted key storage.
- Shipped Docker-sandboxed execution for Python, PPTX, web-app, and office-document workloads — per-task images with enforced memory, CPU, process, output, and wall-clock limits, isolated so sandbox artifacts never touch raw data-source credentials.
- Designed capability-based data intelligence (intent → source discovery → GroundedPlan → SQL compilation → CanonicalResultSet), backed by deterministic temporal interpretation, metric contracts, unit validation, and source provenance.
- Implemented pre-call context-budget enforcement (token estimation, hierarchical history compaction, semantic deduplication, externalization of oversized tool results), eliminating context-overflow failures across differing model context limits.
- Built the evaluation layer: regression scenarios for source selection, temporal interpretation, SQL semantics, authorization, and tool-loop limits; deterministic artifact validation; LLM-as-judge quality gates; and full execution traces covering model calls, tools, retries, latency, tokens, and policy decisions.

Tech: LangGraph · FastAPI · React · PostgreSQL · Redis · MinIO · Docker · MCP · OpenTelemetry · pytest · GitHub Actions

Software Automation Engineer — ProFabx

Jun 2024 – Sep 2024

Ningbo, China

- Programmed custom Python drivers against the Autodesk Inventor API (win32com.client) to automate dynamic 3D CAD modeling, cutting engineering iteration time for custom fabrications.
- Built a lightweight web interface (JavaScript, Python) enabling real-time parametric model updates driven directly by client specification changes.

Tech: Python · Autodesk Inventor API · win32com.client · JavaScript · Fusion 360

PUBLICATIONS

Peer-reviewed

M. Islam, M. Xue, T. Afra. **Exploring Psychologist-Applied Biomarkers in Bipolar Disorder: A Systematic Framework.** Springer LNCS — Entertainment Computing (ICEC 2025), Tokyo, Japan. pp. 61–75. doi:10.1007/978-3-032-02534-0_8

T. Afra, M. Xue, M. Islam. **Enhancing Biofeedback Interventions for Depression and Anxiety Through Entertainment Computing: A Systematic Review.** Springer LNCS — Entertainment Computing (ICEC 2025), Tokyo, Japan. pp. 37–51. doi:10.1007/978-3-032-02534-0_6

Conference presentations (non-archival)

M. Islam, M. Xue, T. Afra. **BEPSBot: A Dual-Mode Writing Assistant for Peer Support in Bipolar Disorder Communities.** HHME 2026 PCC, Oral Presentation, Zhejiang University. N=24 within-subjects study; suggestion adoption 33.3% → 75.6%; SUS 86.98.

T. Afra, M. Xue, M. Islam. **How Should Voice Agents Respond to Anger and Sadness? A Comparison of Empathic Response, Affect-Neutral Acknowledgment, and Cognitive Reappraisal.** HHME 2026 PCC, Oral Presentation, Zhejiang University.

EDUCATION

M.Eng. in Industrial Design Engineering — Zhejiang University

Sep 2023 – Dec 2026 (Expected) · Zhejiang, China

B.Eng. in Computer Science and Technology — Yunnan University

Sep 2019 – Jun 2023 · Kunming, China

CERTIFICATIONS

IBM x 3 · DeepLearning.AI x 1 (2025) — LLM lifecycle, PEFT/LoRA & RLHF, transformer fine-tuning, and containerized generative-AI applications.

AWARDS

United Nations Technology Bank — Global Youth Talent Program, 2023. Selected for an international design-education initiative focused on technology, design, and sustainable development.